NOTRE DAME UNIVERSITY BANGLADESH

Operating System Lab Report

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Course Title: Operating System Lab
Lab Report: Lab Task- 04, 05, 06, 07, 08

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Table of Contents

4 Lab Report-04 : CPU Scheduling **1**
4.1 Implementation of FCFS Scheduling 1
4.2 Implementation of Shortest Job First (SJF) 4

5 Lab Report-05 : Deadlock - Bankers Algorithm **6**
5.1 Implementation of Bankers Algorithm 6

6 Lab Report-06 : Dynamic Storage-Allocation Problem **8**
6.1 Dynamic Storage-Allocation using FirstFit 8
6.2 Dynamic Storage-Allocation using BestFit 10
6.3 Dynamic Storage-Allocation using WorstFit 12

7 Lab Report-07 : Page Replacement Algorithms **14**
7.1 Implementation of Page Replacement using FIFO Algorithm 14
7.2 Implementation of Optimal Algorithm 16
7.3 Page Replacement using LRU 18
7.4 Implementation of Page fitting (hit/miss) 20

8 Lab Report-08 : Disk Scheduling Algorithms **22**
8.1 Disk scheduling using FCFS Algorithm 22
8.2 Disk scheduling using SSTF Algorithm 24
8.3 Disk scheduling using SCAN Algorithm 26

4 Lab Report-04 : CPU Scheduling

4.1 Implementation of FCFS Scheduling

• **Code :**

```
#include <bits/stdc++.h>
using namespace std;

int main()
{
    //cout << "Enter Number of Process : ";
    int n=3;
    //cin >> n;

    int i, j, CT[n], TAT[n], WT[n];
    int PS[]={1,2,3};
    int AT[]={1,2,3};
    int BT[]={24,3,3};
   /* for(i=1; i<=n; i++)
    {
        cout << "Enter Arrival Time for P" << i << ": ";
        cin >> AT[i];
        cout << "Enter Burst Time for P" << i << ": ";
        cin >> BT[i];
    } */
    cout<< endl;
    for(i=0; i<n-1; i++)
    {
        for(j=0; j<n-i-1; j++)
        {
            if(AT[j] > AT[j+1])
            {

                swap(AT[j],AT[j+1]);
            }
        }
    }

    CT[0]=AT[0]+BT[0];
    for(i=1; i<n; i++)
    {
        CT[i]=CT[i-1]+BT[i];
    }

    int Total_T, Total_W;
    for(i=0; i<n; i++)
    {
        TAT[i]=CT[i]-AT[i];
        WT[i]=TAT[i]-BT[i];
```

```
        Total_T=Total_T+TAT[i];
        Total_W=Total_W+WT[i];
    }

    cout << "Process\tArrival time\tBurst time\tCompletition time\tTurnaround ti

    for (i=0; i<n; i++)
    {
        cout << "P"<<PS[i]<<"\t"<<AT[i]<<"\t"<<BT[i]<<"\t"<<CT[i]<<"\t"<<TAT[i]<
    }

    cout << "Average Turnaround Time : " <<Total_T/n<<endl;
    cout << "Average Waiting Time : " <<Total_W/n<<endl;
}
```

- **Input / Output :**

Figure 1: Enter Caption

- **Description :**

4.2 Implementation of Shortest Job First (SJF)

• **Code :**

```
\documentclass{article}
\begin{document}
\title{Practice Lab-01 :}
\author{Istiak Alam}
\maketitle
In March 2006, Congress raised that ceiling an additional \$0.79 trillion
to \$8.97 trillion, which is approximately 68\% of GDP. As of October 4,
2008, the ``Emergency Economic Srabilization Act of 2008" raised the
current debt ceiling to \$11.3 trillion.
\end{document}
```

• **Input / Output :**

- **Description :**

5 Lab Report-05 : Deadlock - Bankers Algorithm

5.1 Implementation of Bankers Algorithm

• **Code :**

```
\documentclass{article}
\begin{document}
\title{Practice Lab-01 :}
\author{Istiak Alam}
\maketitle
In March 2006, Congress raised that ceiling an additional \$0.79 trillion
to \$8.97 trillion, which is approximately 68\% of GDP. As of October 4,
2008, the ``Emergency Economic Srabilization Act of 2008" raised the
current debt ceiling to \$11.3 trillion.
\end{document}
```

• **Input / Output :**

- **Description :**

6 Lab Report-06 : Dynamic Storage-Allocation Problem

6.1 Dynamic Storage-Allocation using FirstFit

• **Code :**

```
\documentclass{article}
\begin{document}
\title{Practice Lab-01 :}
\author{Istiak Alam}
\maketitle
In March 2006, Congress raised that ceiling an additional \$0.79 trillion
to \$8.97 trillion, which is approximately 68\% of GDP. As of October 4,
2008, the ``Emergency Economic Srabilization Act of 2008" raised the
current debt ceiling to \$11.3 trillion.
\end{document}
```

• **Input / Output :**

- **Description :**

6.2 Dynamic Storage-Allocation using BestFit

• **Code :**

```
\documentclass{article}
\begin{document}
\title{Practice Lab-01 :}
\author{Istiak Alam}
\maketitle
In March 2006, Congress raised that ceiling an additional \$0.79 trillion
to \$8.97 trillion, which is approximately 68\% of GDP. As of October 4,
2008, the ``Emergency Economic Srabilization Act of 2008" raised the
current debt ceiling to \$11.3 trillion.
\end{document}
```

• **Input / Output :**

- **Description :**

6.3 Dynamic Storage-Allocation using WorstFit

• **Code :**

```
\documentclass{article}
\begin{document}
\title{Practice Lab-01 :}
\author{Istiak Alam}
\maketitle
In March 2006, Congress raised that ceiling an additional \$0.79 trillion
to \$8.97 trillion, which is approximately 68\% of GDP. As of October 4,
2008, the ``Emergency Economic Srabilization Act of 2008" raised the
current debt ceiling to \$11.3 trillion.
\end{document}
```

• **Input / Output :**

- **Description :**

7 Lab Report-07 : Page Replacement Algorithms

7.1 Implementation of Page Replacement using FIFO Algorithm

• **Code :**

```
\documentclass{article}
\begin{document}
\title{Practice Lab-01 :}
\author{Istiak Alam}
\maketitle
In March 2006, Congress raised that ceiling an additional \$0.79 trillion
to \$8.97 trillion, which is approximately 68\% of GDP. As of October 4,
2008, the ``Emergency Economic Srabilization Act of 2008" raised the
current debt ceiling to \$11.3 trillion.
\end{document}
```

• **Input / Output :**

- **Description :**

7.2 Implementation of Optimal Algorithm

• **Code :**

```
\documentclass{article}
\begin{document}
\title{Practice Lab-01 :}
\author{Istiak Alam}
\maketitle
In March 2006, Congress raised that ceiling an additional \$0.79 trillion
to \$8.97 trillion, which is approximately 68\% of GDP. As of October 4,
2008, the ``Emergency Economic Srabilization Act of 2008" raised the
current debt ceiling to \$11.3 trillion.
\end{document}
```

• **Input / Output :**

- **Description :**

7.3 Page Replacement using LRU

• **Code :**

```
\documentclass{article}
\begin{document}
\title{Practice Lab-01 :}
\author{Istiak Alam}
\maketitle
In March 2006, Congress raised that ceiling an additional \$0.79 trillion
to \$8.97 trillion, which is approximately 68\% of GDP. As of October 4,
2008, the ``Emergency Economic Srabilization Act of 2008" raised the
current debt ceiling to \$11.3 trillion.
\end{document}
```

• **Input / Output :**

- **Description :**

7.4 Implementation of Page fitting (hit/miss)

• **Code :**

```
\documentclass{article}
\begin{document}
\title{Practice Lab-01 :}
\author{Istiak Alam}
\maketitle
In March 2006, Congress raised that ceiling an additional \$0.79 trillion
to \$8.97 trillion, which is approximately 68\% of GDP. As of October 4,
2008, the ``Emergency Economic Srabilization Act of 2008" raised the
current debt ceiling to \$11.3 trillion.
\end{document}
```

• **Input / Output :**

- **Description :**

8 Lab Report-08 : Disk Scheduling Algorithms

8.1 Disk scheduling using FCFS Algorithm

• **Code :**

```
\documentclass{article}
\begin{document}
\title{Practice Lab-01 :}
\author{Istiak Alam}
\maketitle
In March 2006, Congress raised that ceiling an additional \$0.79 trillion
to \$8.97 trillion, which is approximately 68\% of GDP. As of October 4,
2008, the ``Emergency Economic Srabilization Act of 2008" raised the
current debt ceiling to \$11.3 trillion.
\end{document}
```

• **Input / Output :**

- **Description :**

8.2 Disk scheduling using SSTF Algorithm

• **Code :**

```
\documentclass{article}
\begin{document}
\title{Practice Lab-01 :}
\author{Istiak Alam}
\maketitle
In March 2006, Congress raised that ceiling an additional \$0.79 trillion
to \$8.97 trillion, which is approximately 68\% of GDP. As of October 4,
2008, the ``Emergency Economic Srabilization Act of 2008" raised the
current debt ceiling to \$11.3 trillion.
\end{document}
```

• **Input / Output :**

- **Description :**

8.3 Disk scheduling using SCAN Algorithm

• **Code :**

```
\documentclass{article}
\begin{document}
\title{Practice Lab-01 :}
\author{Istiak Alam}
\maketitle
In March 2006, Congress raised that ceiling an additional \$0.79 trillion
to \$8.97 trillion, which is approximately 68\% of GDP. As of October 4,
2008, the ``Emergency Economic Srabilization Act of 2008" raised the
current debt ceiling to \$11.3 trillion.
\end{document}
```

• **Input / Output :**

- **Description :**